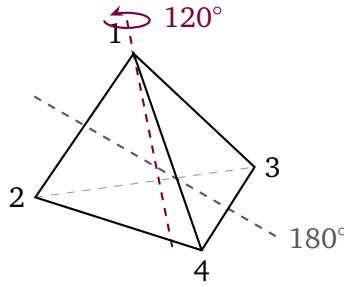


## Symmetries in Physics — Exercise Sheet 5

### Characters and character tables

#### Variant: Questions only

**Exercise 1.** Construct the character table of  $A_4$ , the rotation group of the tetrahedron. The picture, taken from Lecture 2, shows one rotation axis of each kind:



Here is everything you will need (Lecture 2's example  $A_4/V \cong C_3$ ).

- $|A_4| = 12$ , and  $A_4$  has four conjugacy classes: the identity; the three half-turns  $(1\ 2)(3\ 4)$ ,  $(1\ 3)(2\ 4)$ ,  $(1\ 4)(2\ 3)$ ; the four 3-cycles  $(1\ 2\ 3)$ ,  $(1\ 3\ 4)$ ,  $(1\ 4\ 2)$ ,  $(2\ 4\ 3)$ ; and the four inverse 3-cycles  $(1\ 3\ 2)$ ,  $(1\ 4\ 3)$ ,  $(1\ 2\ 4)$ ,  $(2\ 3\ 4)$ .
  - The identity and the three half-turns form a normal subgroup  $V$ , and  $A_4/V \cong C_3$ . The projection  $\pi: A_4 \rightarrow C_3$  sends  $V$  to  $e$ , the class of 3-cycles containing  $(1\ 2\ 3)$  to  $r$ , and the class of  $(1\ 3\ 2)$  to  $r^2$ .
  - The characters of  $C_3$  are  $\chi_k(r^m) = \omega^{km}$ , with  $\omega = e^{2\pi i/3}$  and  $k = 0, 1, 2$ .
- (i) Without constructing any representation, use the two counting theorems of Lecture 5 to find the dimensions of the irreducible representations of  $A_4$ . Do the same for  $D_6$ , which has order 12 and six conjugacy classes (Sheet 2, Exercise 2).
- (ii) Construct three one-dimensional representations of  $A_4$  as  $\chi_k \circ \pi$ .
- (iii) Find the missing row of the table from  $\sum_i n_i \chi_i(g) = 0$  for  $g \neq e$ , and check its norm.

**Exercise 2.** Lecture 5's example on  $D_4$  claimed that the quaternion group  $Q_8$  has the same character table. Prove it. Here is everything you will need (Sheet 4, Exercise 6).

- $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ , with  $i^2 = j^2 = k^2 = ijk = -1$ . Any two of  $i, j, k$  anticommute, for instance  $ij = k = -ji$ .
- Four one-dimensional representations  $\psi_{ab}$ , with  $a, b = \pm 1$ :

$$\psi_{ab}(\pm 1) = 1, \quad \psi_{ab}(\pm i) = a, \quad \psi_{ab}(\pm j) = b, \quad \psi_{ab}(\pm k) = ab.$$

- A two-dimensional irreducible representation

$$i \mapsto -i\sigma_1, \quad j \mapsto -i\sigma_2, \quad k \mapsto -i\sigma_3, \quad -1 \mapsto -1.$$

Here  $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ ,  $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  are the Pauli matrices.

- The table of  $D_4$  (Lecture 5):

| $D_4$ | $1\ e$ | $1\ r^2$ | $2\ r$ | $2\ s$ | $2\ rs$ |
|-------|--------|----------|--------|--------|---------|
| $A_1$ | 1      | 1        | 1      | 1      | 1       |
| $A_2$ | 1      | 1        | 1      | -1     | -1      |
| $B_1$ | 1      | 1        | -1     | 1      | -1      |
| $B_2$ | 1      | 1        | -1     | -1     | 1       |
| $E$   | 2      | -2       | 0      | 0      | 0       |

The lecture labels these rows  $\chi_1, \dots, \chi_5$ ; here we use Mulliken's symbols.

- Find the conjugacy classes of  $Q_8$  and their sizes.
- Prove that these five representations are all the irreducible representations of  $Q_8$ .
- Write down the character table of  $Q_8$  and compare it with that of  $D_4$ .
- Prove that  $D_4 \not\cong Q_8$ . What does this say about the character table?

**Exercise 3.**  $S_4$  permutes the coordinates of  $\mathbb{C}^4$ . Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

**Exercise 4.** Let  $\rho$  be a  $d$ -dimensional representation of a finite group  $G$ , and  $\chi$  its character.

- Show that  $|\chi(g)| \leq d$  for every  $g$ , with equality if and only if  $\rho(g)$  is a scalar multiple of the identity. Deduce that  $\chi(g) = d$  exactly on the kernel of  $\rho$ .
- Suppose that  $\rho$  is faithful. Prove that  $|\chi(g)| = d$  only for central elements  $g$ .
- From the tables of Exercise 2, find which irreducible representations of  $D_4$  and  $Q_8$  are faithful, and at which elements  $|\chi(g)| = d$ .